UI/UX Evaluation Methods Report

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Abstract

This report is written for the ICT Project 2026 module at Lab University of Applied Sciences. It reviews the UI/UX evaluation methods from the course material and selects the most suitable method from each of the five method of ui-ux design, documents for the Lindstorm Warehouse Voice Picking System. For every chosen method, the report explains what the method is, why it fits this project, and why the other methods in the same document were not selected.

The Lindstorm system is a hands-free, voice-guided warehouse picking application. Workers open Chrome on their phone, log in, and follow spoken instructions to pick products from shelf locations without typing or touching the screen. Because voice is the main input channel, some common UX methods cannot be used as they would interfere with the microphone.

1. Introduction

The Lindstorm Warehouse Voice Picking System was built as the main deliverable for the ICT Project 2026 module. It is a browser-based application that lets warehouse workers complete their picking tasks hands-free using voice commands on a mobile phone.

The system uses Python Flask for the backend, SQLite as the database, and the Web Speech API in Chrome for voice input and output. The worker hears a spoken instruction, walks to the shelf, speaks a confirmation code, and says 'done' after picking the items. The supervisor creates orders from a separate dashboard and monitors progress live.

Choosing the right UI/UX evaluation methods requires care. Because voice is the primary interaction mode, methods like Think-Aloud Protocol cannot be used -they require the user to speak their thoughts aloud, which would confuse the speech recognition and break the picking session. This report selects one method from each of the four course PDFs, explains why it is the best match, and notes why the other methods in the same PDF were not chosen.

2. Method from PDF A — Accessibility Testing

What is Accessibility Testing?

Accessibility testing checks whether a system can be used by people with different physical, cognitive, and sensory abilities. It follows guidelines such as WCAG (Web Content Accessibility Guidelines), which cover text readability, colour contrast, keyboard navigation, clear labels, and meaningful error messages.

Why it suits this project

The Lindstorm system is used in a real warehouse environment where workers may have different needs. Some may not be native speakers, which affects how they understand voice prompts. Some may have difficulty speaking clearly, affecting speech recognition. Others may have hearing difficulties, meaning the text on screen must be large and readable without relying on audio alone.

A formal accessibility review would check:

- Whether the confirmation code and quantity text meet contrast requirements against thebackground.
- Whether the keyboard fallback input is always reachable and clearly labelled for workers whocannot use voice.
- Whether error messages are clear enough for users who missed the audio feedback.
- Whether the microphone button has a proper accessible label.

The project already made several accessibility-aware decisions 56px location text, 96px quantity text, and a keyboard fallback at all times. A formal review would confirm what works and identify what still needs improvement.

Why the other PDF A methods were not chosen

A/B Testing needs large numbers of real users split into groups, which is not possible for a student project with two demo accounts. Data Analytics requires weeks of genuine warehouse usage, not demo data. Eye-Tracking needs specialist equipment and is most useful for visually complex screens -the worker screen in this system is intentionally minimal. User Testing is a strong method in general but cannot use Think-Aloud Protocol here because the microphone is always listening, and recruiting real warehouse workers was not feasible.

3. Method from PDF B — Heuristic Evaluation

What is Heuristic Evaluation?

Heuristic Evaluation is an expert-based method where an evaluator reviews an interface against a set of recognised usability principles called heuristics. The most well-known set is Nielsen's 10 Heuristics, covering principles such as visibility of system status, error prevention, consistency, and help and documentation.

Why it suits this project — and how it was applied

Heuristic Evaluation was chosen as the primary evaluation method and was already carried out during development. It was the right choice because it can be done by a single evaluator without recruiting participants, it works on a fully built system, and it produces clear findings that can be directly turned into improvements.

The evaluation found four issues and led to four fixes:

- H10 — Help was missing: a help modal with step-by-step instructions was added to the workernavigation bar.
- H9 — Error recovery was poor: wrong confirmation codes looped. The system now speaks theerror once and stops.
- H6 — Recognition over recall: location text was increased to 56px and quantity to 96px soworkers can read while walking.
- H7 — Flexibility: the supervisor has no bulk import option. This was recorded as a futureimprovement.

These results show that Heuristic Evaluation was genuinely useful -it produced real changes to the system, not just theoretical recommendations.

Why the other PDF B methods were not chosen

Design Sprint is a five-day team workshop for early-stage design problems. This project is built by one student and is already complete, so a sprint is not applicable. AttrakDiff measures emotional and aesthetic appeal -useful for consumer apps but not for a functional workplace tool. The Trunk Test checks navigation orientation, but the worker screen has almost no navigation to test.

4. Method from PDF C — Guerrilla Testing

What is Guerrilla Testing?

Guerrilla Testing is an informal, low-cost usability method where feedback is collected quickly from people in everyday settings such as hallways or cafeterias. Participants are approached informally and asked to try one or two tasks. The goal is to catch obvious usability problems early without needing a formal lab or scheduled sessions.

Why it suits this project

Guerrilla Testing is the most realistic form of user feedback available for a solo student project. A classmate or university staff member can be given a phone running the Lindstorm system and asked to try the picking flow from scratch. Because they have no knowledge of how the system works, they represent exactly the right condition -a first-time warehouse worker.

Guerrilla Testing would reveal things like:

- Whether the spoken instruction is clear enough the first time a user hears it.
- Whether the user understands what the microphone button does before tapping it.
- Whether the error message makes sense when the wrong code is spoken.
- Whether the transition between tasks feels smooth or confusing.

These are exactly the kinds of problems that expert evaluation alone cannot catch , they only appear when someone unfamiliar with the system uses it for the very first time.

Why the other PDF C methods were not chosen

Crazy 8's is an ideation method used before building - it is not an evaluation tool for a finished system. UX Prototyping was done during development but applying it formally now would mean building a mock-up of something that already works. Six Thinking Hats is a group facilitation method and loses value when used alone. The Five-Second Test checks first visual impressions, but the worker screen is so simple it would pass trivially and reveal nothing useful.

5. Method from PDF D — User Flows

What are User Flows?

User Flows are visual diagrams that map every step a user takes to complete a task. They show the starting point, each interaction or decision along the way, possible error paths, and the end result. They make the interaction logic of a system visible and easy to review.

Why it suits this project

The Lindstorm system has two different user journeys -the worker picking flow and the supervisor order management flow and both benefit from being mapped out clearly.

The worker flow involves these steps:

- Open Chrome on phone and go to the server address.
- Log in with username and password.
- Hear the first voice instruction — location and confirmation code.
- Walk to the shelf and tap the microphone button.
- Speak the confirmation code — system checks it against the database.
- If correct: hear the quantity and say 'done' after picking.
- If wrong: hear the error message once and try again.
- Receive the next location instruction automatically.
- When all tasks are done, wait — system polls every 4 seconds for new orders.

Drawing this as a flow diagram makes it easy to spot edge cases for example, what happens if a new task arrives exactly when the worker finishes the last one. User Flows also serve as useful documentation for anyone maintaining or extending the system in the future.

Why the other PDF D methods were not chosen

Concept Testing is used before building a system to validate the idea it is too early-stage for a completed project. Kano Analysis helps teams prioritise which features to build next, but the project scope is already decided. Jobs to Be Done confirms the job the system was already designed around, so it does not produce new evaluation findings. Card Sorting helps design navigation structures, but the worker screen has no navigation and the supervisor dashboard has only a few clearly separated tabs.

6. Summary

PDF	Chosen Method	Status	Main Reason
A	Accessibility Testing	Recommended	Voice apps have unique accessibility needs; WCAG covers audio, contrast, and keyboard fallback
B	Heuristic Evaluation	Applied — Primary	Expert-based, solo-applicable, produced four real UI fixes during development
C	Guerrilla Testing	Recommended	No lab needed; tests unfamiliar users on the core voice flow quickly and cheaply
D	User Flows	Recommended	Maps both user journeys, shows edge cases, documents the voice dialogue sequence

It is also worth noting that Cognitive Walkthrough was used as the secondary evaluation method during development. The evaluator stepped through each task and asked whether a first-time user would understand what to do at each point - especially important for the voice dialogue, which must be learnable without any prior training.

7. Conclusion

The Lindstorm Warehouse Voice Picking System is different from a typical web application because voice is the main way users interact with it. This makes some common UX evaluation methods unsuitable and requires careful thinking about which methods actually fit the context.

Heuristic Evaluation was the right primary method because it was practical for a solo developer, could be applied to the finished system, and produced real improvements. Cognitive Walkthrough worked well as a secondary method to think through the worker's experience step by step.

Accessibility Testing is the most important recommendation going forward. A voice system used in a physical workplace must work for people with different language backgrounds, speech patterns, and sensory abilities. A formal WCAG review would make this clear and actionable.

Guerrilla Testing is the easiest next step. Asking a few unfamiliar people to try the picking flow on a phone would quickly reveal whether the voice instructions are clear and whether the interface makes sense to someone using it for the first time.

User Flows are valuable for documentation. Drawing out the worker and supervisor journeys as diagrams would make the full interaction logic visible, highlight edge cases, and leave useful records for anyone who works on the system later.

Together, these four methods one already applied as primary, one as secondary, and two recommended for future use give a thorough picture of how UI/UX evaluation fits the Lindstorm project.